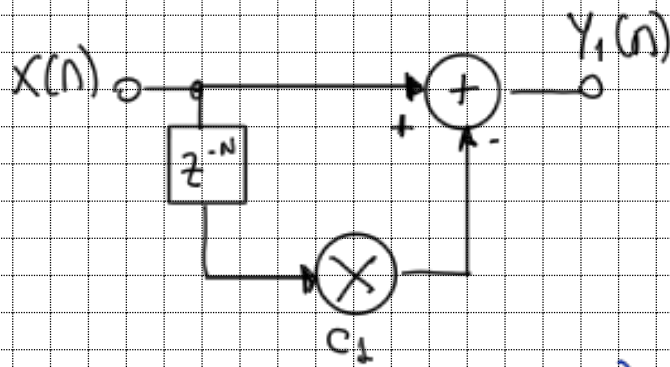


①



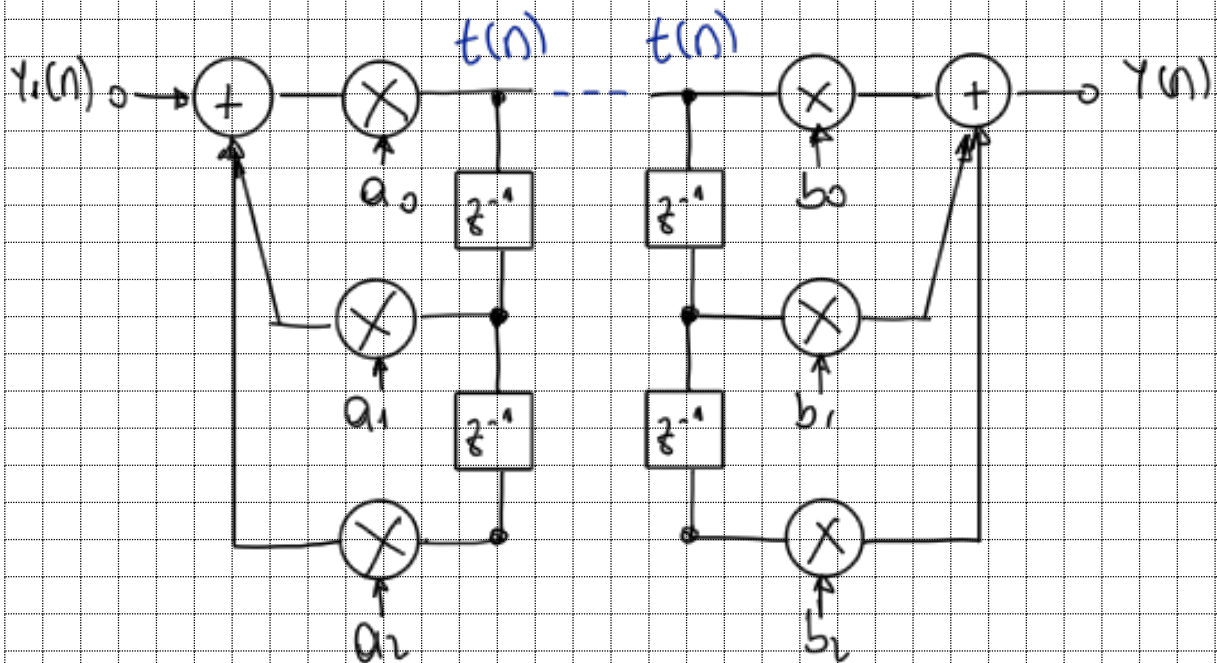
$$y_1(n) = x(n) - C_1 x(n-N)$$

↪ $\begin{pmatrix} 1 \\ z^{-N} \end{pmatrix}$

$$Y_1(z) = X(z) - C_1 z^{-N} X(z)$$

$$Y_1(z) = X(z) (1 - C_1 z^{-N})$$

$$\frac{Y_1(z)}{X(z)} = H_1(z) = 1 - C_1 z^{-N}$$



$$t(n) = a_0(y_1(n) + a_1 t(n-1) + a_2 t(n-2))$$

↪ $T(z) = a_0 (Y_1(z) + a_1 z^{-1} T(z) + a_2 z^{-2} T(z))$

$$T(z) - a_0 a_1 z^{-1} T(z) - a_0 a_2 z^{-2} T(z) = a_0 Y_1(z)$$

$$H_2(z) = \frac{T(z)}{Y_1(z)} = \frac{Q_0}{1 - a_0 a_1 z^{-1} - Q_0 a_2 z^{-2}}$$

$$H_2(z) = \frac{1}{\frac{1}{Q_0} - a_1 z^{-1} - a_2 z^{-2}}$$

$$y(n) = b_0 t(n) + b_1 t(n-1) + b_2 t(n-2)$$

↪

$$Y(z) = b_0 T(z) + b_1 z^{-1} T(z) + b_2 z^{-2} T(z)$$

$$H_3(z) = \frac{Y(z)}{T(z)} = b_0 + b_1 z^{-1} + b_2 z^{-2}$$

$$H(z) = H_1(z) \cdot H_2(z) \cdot H_3(z)$$

$$= \frac{\cancel{Y_1(z)}}{\cancel{X(z)}} \cdot \frac{\cancel{T(z)}}{\cancel{Y_1(z)}} \cdot \frac{Y(z)}{\cancel{T(z)}} = \frac{Y(z)}{X(z)} = H(z)$$

$$H(z) = (1 - c_1 z^{-N}) \frac{b_0 + b_1 z^{-1} + b_2 z^{-2}}{\frac{1}{Q_0} - a_1 z^{-1} - a_2 z^{-2}}$$

② Filtro de media móvil

$$a_0 = a_1 = 1, \quad b_0 = \frac{1}{N}, \quad c_1 = 1$$

$$H(z) = (1 - c_1 z^{-N}) \frac{b_0 + b_1 z^{-1} + b_2 z^{-2}}{\frac{1}{a_0} - a_1 z^{-1} - a_2 z^{-2}}$$

$$H(z) = (1 - z^{-N}) \cdot \frac{\frac{1}{N} + b_1 z^{-1} + b_2 z^{-2}}{1 - z^{-1} - a_2 z^{-2}}$$

$$= \left(1 - \frac{1}{z^N}\right) \cdot \frac{\frac{1}{N} + \frac{b_1}{z} + \frac{b_2}{z^2}}{1 - \frac{1}{z} + \frac{a_2}{z^2}}$$

$$= \frac{z^N - 1}{z^N} \cdot \frac{\frac{z^2 + b_1 z N + b_2 N}{N z^2}}{\frac{z^2 - z + a_2}{z^2}}$$

$$\frac{z^N - 1}{z^N} \cdot \frac{z^2 + b_1 z N + b_2 N}{N(z^2 - z + a_2)}$$

el filtro media móvil, tiene la forma

$$y(n) = \frac{1}{N} \sum_{k=0}^{N-1} x(n-k)$$

$$n=1 \Rightarrow y(1) = \frac{1}{1} (x(1-0)) = x(1)$$

$$n=2 \Rightarrow y(2) = \frac{1}{2} (x(2-0) + x(2-1))$$

$$y(2) = \frac{1}{2} (x(2) + (1))$$

verif. ca

$$y(n) = \frac{1}{N} \sum_{k=0}^{N-1} x(n-k)$$

$$y(z) = \frac{1}{N} \sum_{k=0}^{N-1} x(n) z^{-k}$$

$$y(z) = \frac{x(z)}{N} \sum_{k=0}^{N-1} z^{-k} \rightarrow \frac{y(z)}{x(z)} = H(z) = \frac{1}{N} \sum_{k=0}^{N-1} z^{-k}$$

$$H(z) = \frac{1}{N} (1 + z^{-1} + z^{-2} + \dots + z^{-(N-1)})$$

$$r = z^{-1} ; r^2 = z^{-2} ; r^{N-1} = z^{-(N-1)}$$

$$S = 1 + r + r^2 + \dots + r^{N-1}$$

$$Sr = r + r^2 + r^3 + \dots + r^N$$

$$S - Sr = 1 - r^N$$

$$S(1-r) = \frac{1-r^N}{1-r} = S \Rightarrow \frac{1-r^N}{1-r} = \frac{1-z^{-N}}{1-z^{-1}}$$

$$\Rightarrow H(z) = \frac{1}{N} \sum_{k=0}^{N-1} z^{-k} = \frac{1}{N} \left(\frac{1-z^{-N}}{1-z^{-1}} \right)$$

Necesito $\downarrow$

$$\frac{z^N - 1}{z^N} \cdot \frac{z^2 + b_1 z N + b_2 N}{N(z^2 - z + a_2)} = \frac{1}{N} \left(\frac{1 - z^{-N}}{1 - z^{-1}} \right)$$

$$(1 - z^{-N}) \left(\frac{z^2 + b_1 N z + b_2 N}{N(z^2 - z + a_2)} \right) = \frac{1}{N} \left(\frac{1 - z^{-N}}{1 - z^{-1}} \right)$$

$$\text{Si } b_1 = b_2 = a_2 = 0$$

$$\Rightarrow \frac{1 - z^{-N}}{N} \cdot \frac{z^2}{z^2 - z} = \frac{1}{N} \left(\frac{1 - z^{-N}}{1 - z^{-1}} \right)$$

$$\frac{1}{N} (1 - z^{-N}) \left(\frac{1}{z^2 - z} \right) = \frac{1}{N} \left(\frac{1 - z^{-N}}{1 - z^{-1}} \right)$$

$$\frac{1}{N} (1 - z^{-N}) \frac{1}{1 - z^{-1}} = \frac{1}{N} \frac{1 - z^{-N}}{1 - z^{-1}}$$

$$\Rightarrow H(z) = \frac{1}{N} \sum_{k=0}^{N-1} z^{-k}$$

$$N=3 \quad H(z) = \frac{1}{3} \sum_{k=0}^{3-1} z^{-k}$$

$$= \frac{1}{3} (1 + z^{-1} + z^{-2}) = \frac{1}{3} \frac{z^2 + z + 1}{z^2}$$

o Si No

$$H(z) = \frac{1}{3} \cdot \frac{(1 - z^{-3})}{1 - z^{-1}} = \frac{1}{3} \cdot \frac{1}{z^3} \frac{(z^3 - 1)}{\frac{z - 1}{z}} = \frac{1}{3} \frac{z^3 - 1}{z^2(z - 1)}$$

$$a^3 - b^3 = (a - b)(a^2 + ab + b^2)$$

$$H(z) = \frac{1}{3} \frac{(z - 1)(z^2 + z + 1)}{(z - 1)z^2}$$

$$H(z) = \frac{1}{3} \left(\frac{z^2 + z + 1}{z^2} \right)$$

polo doble en $z=0$

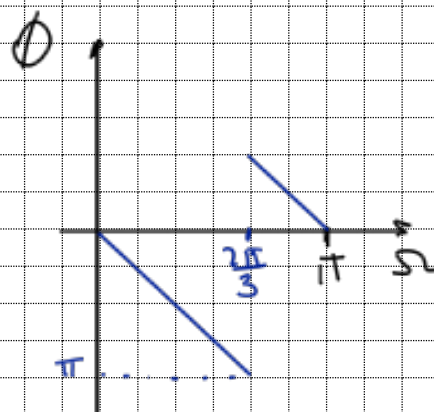
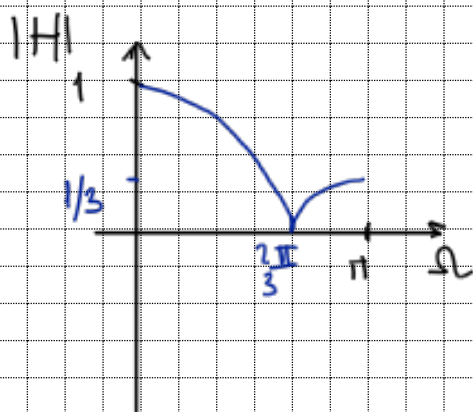
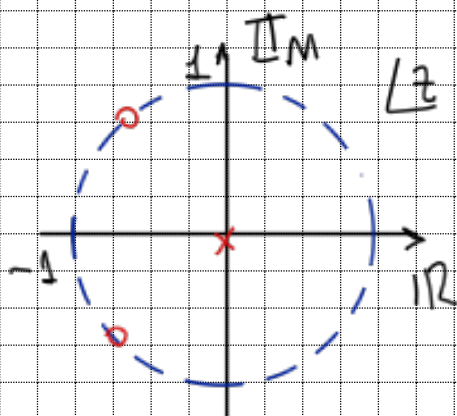
polo simple conjugado en $\frac{1 \pm j\sqrt{3}}{2}$

$$\frac{1}{3} (1 + z^{-1} + z^{-2})$$

$$\frac{1}{3} (e^{j0} + e^{-j\omega} + e^{j\omega})$$

$$\frac{1}{3} (e^{-j\omega} (e^{j\omega} + 1 + e^{j\omega}))$$

$$\frac{1}{3} e^{-j\omega} (1 + 2\cos(\omega))$$



$$N=4$$

$$H(z) = \frac{1}{4} (1 + z^{-1} + z^{-2} + z^{-3}) = \frac{1}{4} \left(\frac{z^3 + z^2 + z + 1}{z^3} \right)$$

$$H(z) = \frac{1}{4} \left(\frac{1 - z^{-4}}{1 - z^{-1}} \right) = \frac{1}{4} \left(\frac{z^4 - 1}{z^4} \right) \cdot \left(\frac{z - 1}{z - 1} \right)$$

$$\frac{1}{4} \frac{1}{z^3} \frac{z^4 - 1}{z - 1} \quad a^4 - b^4 = (a^2 + b^2)(a^2 - b^2)$$

$$\frac{1}{4} \frac{1}{z^3} \frac{(z^2 + 1)(z^2 - 1)}{z - 1} = \frac{1}{4z^3} \frac{(z^2 + 1)(z + 1)(z - 1)}{z - 1}$$

$$\frac{1}{4} \frac{z^3 + z^2 + z + 1}{z^3} = \frac{1}{4} (1 + z^{-1} + z^{-2} + z^{-3})$$

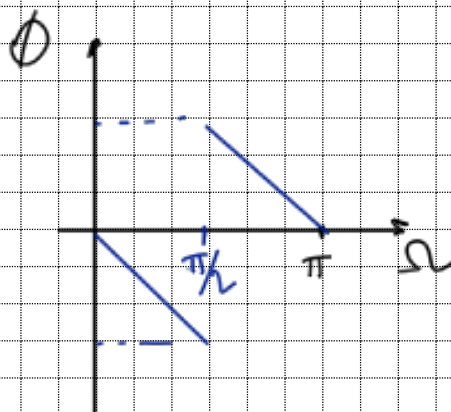
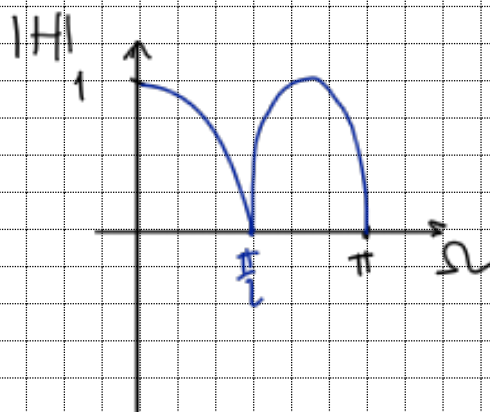
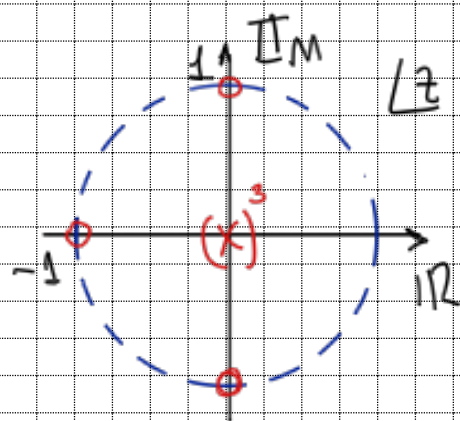
$$\frac{1}{4} (e^{j0} + e^{-j\Omega} + e^{-j2\Omega} + e^{-j3\Omega})$$

$$\Sigma_N = \frac{1}{N} \sum_{k=0}^{N/2} 2b \cos\left((k + \frac{1}{2})\Omega\right) e^{-j\left(\frac{N+1}{2}\right)\Omega}$$

$$\frac{1}{4} \sum_{k=0}^2 2b_{1-k} \cos\left((k + \frac{1}{2})\Omega\right) e^{-j\frac{3}{2}\Omega}$$

$$\left(\frac{1}{2} \cos\left(\frac{\Omega}{2}\right) + \frac{1}{2} \cos\left(\frac{3\Omega}{2}\right) \right) e^{-j\frac{3}{2}\Omega}$$

$$\cos A + \cos B = 2 \cos\left(\frac{A+B}{2}\right) \cos\left(\frac{A-B}{2}\right) \Rightarrow \cos \Omega \cos \frac{\Omega}{2} = 0 \quad \begin{cases} \Omega = \frac{\pi}{2} \\ \Omega = \pi \end{cases}$$



$$N=5$$

$$H(z) = \frac{1}{5} \sum_{k=0}^{5-1} z^{-k}$$

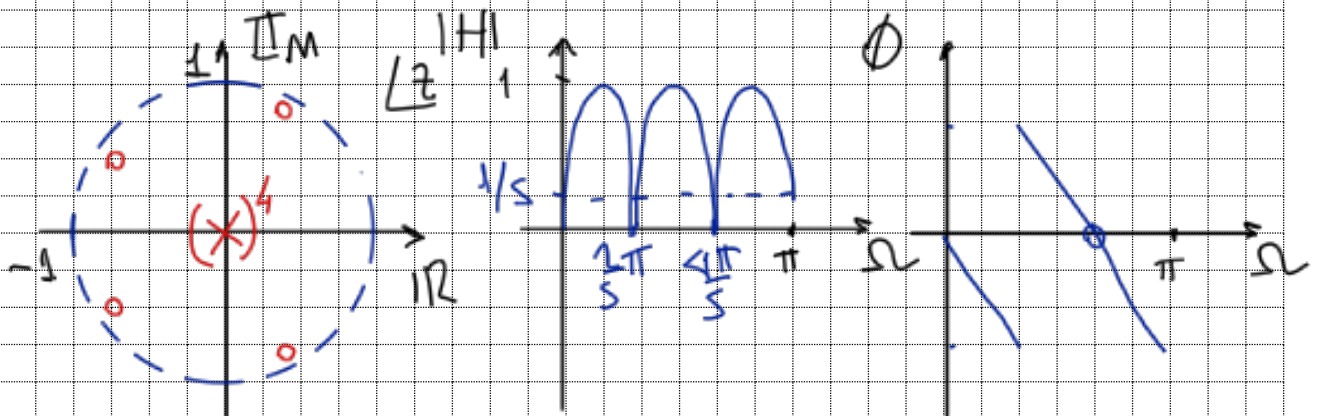
$$T_{II} = \frac{1}{N} \left(1 + \sum_{k=1}^{N-1} 2b_{\frac{N-1}{2}-k} \cos(k\Omega) \right) e^{-j\frac{(N-1)}{2}\Omega}$$

$$H(z) = \frac{1}{5} (1 + z^{-1} + z^{-2} + z^{-3} + z^{-4}) = \frac{1}{5} \left(\frac{1 - z^{-5}}{1 - z^{-1}} \right)$$

$$T(\Omega) = \frac{1}{5} \left(1 + \sum_{k=1}^2 2b_{2-k} \cos(k\Omega) \right) e^{-j2\Omega}$$

$$T(\Omega) = \frac{1}{5} (1 + 2\cos(\Omega) + 2\cos(2\Omega)) e^{-j2\Omega}$$

$$\begin{aligned} \cos(2\alpha) &= \cos^2\alpha - \sin^2\alpha \quad \{ 1 + 2\cos\Omega + 2(2\cos^2\Omega - 1) = 0 \\ \cos(2\alpha) &= 2\cos^2\alpha - 1 \} \quad 4\cos^2\Omega + 2\cos\Omega - 1 = 0 \end{aligned}$$



- "Es un filtro FIR", la respuesta al impulso tiene N muestras

$$H(z) = \frac{1}{N} \sum_{k=0}^{N-1} z^{-k}$$

$$\frac{1}{3} \left(\frac{1 - z^{-3}}{1 - z^{-1}} \right) = \frac{1}{3} (1 + z^{-1} + z^{-2})$$

- Para el filtro media móvil 3 No hay grandes ventajas, si bien, tienen el mismo comportamiento. Las ventajas se presentan con N mas grandes, en donde se reduce el costo computacional con el CIC respecto al FIR "común"

- $h(k) = (1, 1, 1, 1, 1, 1, 1)$

$$\frac{Y(z)}{X(z)} = H(z)$$

$$Y(z) = H(z)X(z)$$

$$\left[\frac{Y(z)}{X(z)} \right]^{-1}$$

$$y(n) = h(n) * x(n) = \sum_k h(k) x(n-k) = y(n)$$

$$Y(z) = \sum_k h(k) X(z) z^{-k}$$

$$Y(z) = X(z) \sum_k h(k) z^{-k}$$

$$\frac{Y(z)}{X(z)} = H(z) = \sum_k h(k) z^{-k}$$

Si $h(k) = (1, 1, 1, 1, 1, 1, 1) \rightarrow N=7$

$$H(z) = \sum_{k=0}^{N-1} h(k) \cdot z^{-k}$$

$$H(z) = 1 + z^{-1} + z^{-2} + z^{-3} + z^{-4} + z^{-5} + z^{-6}$$

anteriormente se había demostrado

que $\sum_{n=0}^{N-1} z^{-k} = \frac{1 - z^{-N}}{1 - z^{-1}}$

$$H(z) = 1 + z^{-1} + z^{-2} + z^{-3} + z^{-4} + z^{-5} + z^{-6} = \frac{1 - z^{-7}}{1 - z^{-1}}$$

$$\frac{1 - z^{-7}}{1 - z^{-1}} = (1 - C_1 z^{-N}) \frac{b_0 + b_1 z^{-1} + b_2 z^{-2}}{\frac{1}{a_0} - a_1 z^{-1} - a_2 z^{-2}}$$

para que sean equivalentes, necesito

$$C_1 = b_0 = 1 \quad b_1 = b_2 = a_2 = 0, \quad a_1 = 1 \quad a_0 = 1$$

$$\Rightarrow \frac{1 - z^{-7}}{1 - z^{-1}} = (1 - 1 \cdot z^{-7}) \frac{1 + 0z^{-1} + 0z^{-2}}{\frac{1}{1} - 1z^{-1} - 0z^{-2}}$$

$$\frac{1 - z^{-7}}{1 - z^{-1}} = 1 - 1z^{-7} \cdot \frac{1}{1 - z^{-1}}$$

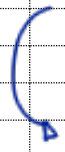
$$\frac{1 - z^{-7}}{1 - z^{-1}} = \frac{1 - z^{-7}}{1 - z^{-1}}$$

se puede con esta topología con

$$C_1 = b_0 = 1 \quad b_1 = b_2 = a_2 = 0, \quad a_1 = 1 \quad a_0 = 1$$

③ diferenciador

$$y(n) = \frac{x(n) - x(n-2]}{2}$$



$\{z\}$

$$Y(z) = \frac{1}{2} (X(z) - X(z)z^{-2})$$

$$Y(z) = \frac{X(z)}{2} (1 - z^{-2}) \Rightarrow \frac{Y(z)}{X(z)} = H(z) = \frac{1}{2} (1 - z^{-2})$$

$$\frac{1}{2} (1 - z^{-2}) = (1 - c_1 z^{-1}) \frac{b_0 + b_1 z^{-1} + b_2 z^{-2}}{1 - a_1 z^{-1} - a_2 z^{-2}}$$

con $N=2$ muestras

$$b_0 = 1/2$$

$$b_1 = b_2 = 0$$

$$a_0 = 1 \quad a_1 = a_2 = 0$$

$$c_1 = 1$$

$$\Rightarrow (1 - 1z^{-2}) \cdot \frac{1/2 + 0z^{-1} + 0z^{-2}}{1 - 0z^{-1} - 0z^{-2}}$$

$$\frac{1 - z^{-2}}{2} = H(z)$$

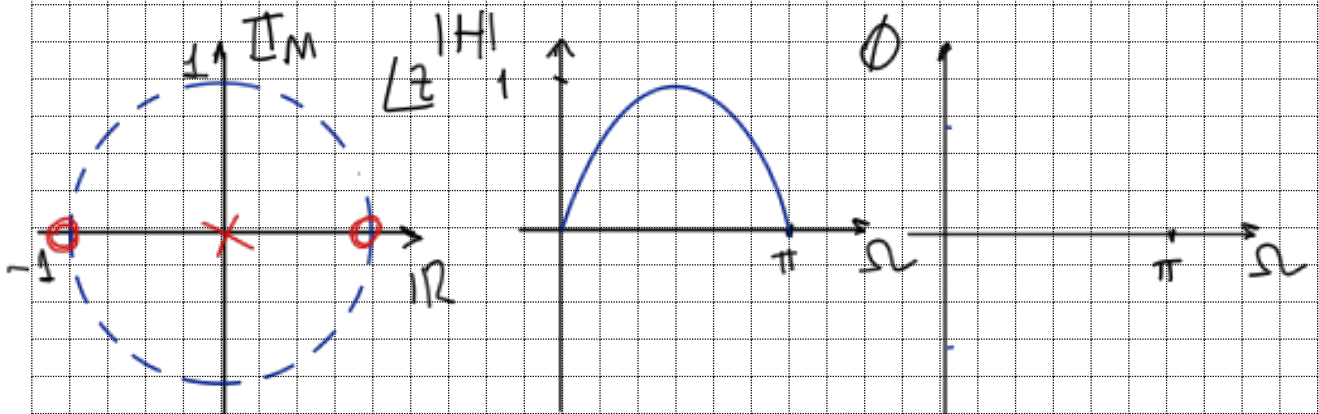
$$b_0 = 1/2$$

$$b_1 = b_2 = 0$$

$$a_0 = 1 \quad a_1 = a_2 = 0$$

$$c_1 = 1$$

$$\frac{z^2 - 1}{2z^2} \Rightarrow \frac{1}{2} e^{-j\Omega} (e^{j\Omega} - e^{-j\Omega}) = \text{sen} \Omega e^{j\frac{3}{2}\Omega}$$



④ $q_0 = b_0 = 1 \quad b_1 = -1 \quad a_1 = \beta \quad , \quad \beta = 0,9$

$$H(z) = (1 - C_1 z^{-N}) \frac{b_0 + b_1 z^{-1} + b_2 z^{-2}}{\frac{1}{q_0} - a_1 z^{-1} - a_2 z^{-2}}$$

$$H(z) = (1 - C_1 z^{-N}) \frac{(1 - z^{-1} + b_2 z^{-2})}{\frac{1}{1} - 0,9 z^{-1} - a_2 z^{-2}}$$

como No tengo b_2, c_2, a_2 .

$$H(s) = \frac{S}{S + \omega_c} \quad \rightarrow \quad S = K \frac{z-1}{z+1} \quad \omega_c = \Omega$$

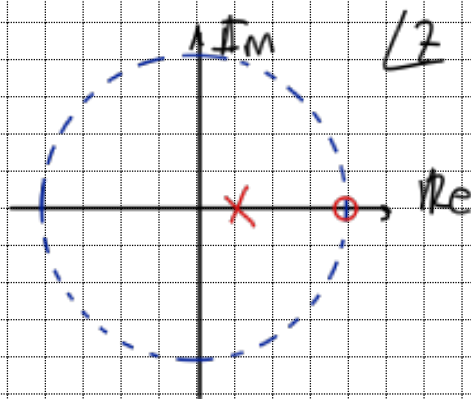
$$\Rightarrow H(z) = \frac{K \frac{z-1}{z+1}}{\frac{K \frac{z-1}{z+1} + \Omega}{z+1}} = \frac{K \frac{(z-1)}{z+1}}{\frac{K(z-1) + \Omega(z+1)}{z+1}} \quad K = 2f_s$$

$$H(z) = \frac{K(z-1)}{K(z-1) + \Omega(z+1)}$$

$$H(z) = \frac{K(z-1)}{z(K+\Omega) + \Omega - K}$$

$$H(z) = \frac{K}{K+\Omega} \frac{z-1}{z + \frac{\Omega-K}{K+\Omega}}$$

$$z = 1 e^{j\Omega} \quad 1 = e^{j0}$$



polo $\frac{\Omega-K}{K+\Omega}$, $K > 0 \wedge \Omega > 0$

el polo se mueve
dentro de la abscisa
entre $[-1; 1]$

$$(1 - c_1 z^{-N}) \frac{(1 - z^{-1} + b_2 z^{-2})}{1 - 0,9 z^{-1} - 0,2 z^{-2}} = \frac{K}{K + \Omega} \frac{z-1}{z + \frac{\Omega-K}{K+\Omega}}$$

$c_1 = b_2 = 0,2 = 0$ para primer orden

$$(1 - 0 z^{-N}) \frac{(1 - z^{-1} + 0 z^{-2})}{1 - 0,9 z^{-1} - 0 z^{-2}}$$

$$\frac{z-1}{z} \frac{1}{z-0,9} = \frac{z-1}{z-0,9} \frac{K}{K+\Omega} = \frac{\Omega-K}{K+\Omega} = -0,9$$

$$\Omega - K = -0,9(K + \Omega)$$

$$\Omega + 0,9\Omega = -0,9K + K$$

$$1,9\Omega = 0,1K \Rightarrow K = 19\Omega$$

$$H(z) = \frac{z-1}{z-0,9}, \text{ con } c_1 = b_2 = 0,2 = 0$$

- $20 \log(|H(\Omega = \pi)|) = A$
 $20 \log(|H(\Omega = 0,1\pi)|) = A - 3 \text{ dB}$

$$20 \log(|H(\Omega = 0,1\pi)|) = 20 \log(|H(\Omega = \pi)|) - 3 \text{ dB}$$

$$20 \log(|H(\Omega = 0,1\pi)|) - 20 \log(|H(\Omega = \pi)|) = -3 \text{ dB}$$

$$20 (\log(|H(\Omega = 0,1\pi)|) - \log(|H(\Omega = \pi)|)) = -3 \text{ dB}$$

$$\log \left(\frac{|H(\Omega = 0,1\pi)|}{|H(\Omega = \pi)|} \right) = \frac{-3}{20}$$

$$\frac{|H(\Omega = 0,1\pi)|}{|H(\Omega = \pi)|} = 10^{\frac{-3}{20}} = \frac{\sqrt{2}}{2}$$

$$|H(\Omega = 0,1\pi)| = \frac{\sqrt{2}}{2} |H(\Omega = \pi)|$$

$$H(z: e^{j\Omega}) = \left| \frac{e^{j\Omega} - 1}{e^{j\Omega} - \beta} \right|$$

$$\frac{\left| \frac{e^{j0,1\pi} - 1}{e^{j0,1\pi} - \beta} \right|}{\left| \frac{e^{j\pi} - 1}{e^{j\pi} - \beta} \right|} = \frac{\sqrt{2}}{2}$$

$$e^{j\alpha\Omega} = \cos(\alpha\Omega) + j\sin(\alpha\Omega)$$

$$\frac{\left| \frac{\cos(0,1\pi) + j\sin(0,1\pi) - 1}{\cos(0,1\pi) + j\sin(0,1\pi) - \beta} \right|}{\left| \frac{\cos(\pi) + j\sin(\pi) - 1}{\cos(\pi) + j\sin(\pi) - \beta} \right|} = \frac{\left| \frac{\cos(0,1\pi) - 1 + j\sin(0,1\pi)}{\cos(0,1\pi) - \beta + j\sin(0,1\pi)} \right|}{\left| \frac{-1 - 1}{-1 - \beta} \right|}$$

$$\frac{\sqrt{\frac{(\cos(0,1\pi) - 1)^2 + \sin^2(0,1\pi)}{(\cos(0,1\pi) - \beta)^2 + \sin^2(0,1\pi)}}}{\left| \frac{-2}{-1 - \beta} \right|} = \frac{\sqrt{2}}{2}$$

$$(\cos(0,1\pi) - 1)^2 = \cos^2(0,1\pi) - 2\cos(0,1\pi) + 1$$

$$(\cos(0,1\pi) - \beta)^2 = \cos^2(0,1\pi) - 2\beta\cos(0,1\pi) + \beta^2$$

$$\cos^2(0,1\pi) + \sin^2(0,1\pi) = 1$$

$$|-2| = 2$$

$$|-1 - \beta| = |(-1)(1 + \beta)| = |1 + \beta| = 1 + \beta$$

$$\frac{\sqrt{\frac{2 - 2\cos(0,1\pi)}{1 + \beta^2 - 2\beta\cos(0,1\pi)}}}{\frac{2}{1 + \beta}} = \frac{\sqrt{\frac{(1 + \beta)^2 (2)(1 - \cos(0,1\pi))}{1 + \beta^2 - 2\beta\cos(0,1\pi)}}}{2} = \frac{\sqrt{2}}{2}$$

$$\sqrt{\frac{(1+\beta^2)(1-\cos(0,1\pi))}{2(1+\beta^2-2\beta\cos(0,1\pi))}} = \frac{\sqrt{2}}{2} \quad c = \cos(0,1\pi)$$

$$\left(\sqrt{\frac{(1+2\beta+\beta^2)(1-c)}{2(1+\beta^2-2\beta c)}} = \frac{\sqrt{2}}{2} \right)^2 \Rightarrow \frac{(1+2\beta+\beta^2)(1-c)}{2(1+\beta^2-2\beta c)} = \cancel{\frac{2}{4}} \frac{1}{2}$$

$$(1+2\beta+\beta^2)(1-c) = 1+\beta^2-2\beta c$$

$$\cancel{1} + \cancel{2\beta} + \cancel{\beta^2} - c - \cancel{2\beta c} - c\beta^2 = \cancel{1} + \cancel{\beta^2} - \cancel{2\beta c}$$

$$0 = c\beta^2 - 2\beta + c$$

$$0 = \beta^2 - \beta \frac{2}{c} + 1$$

$$\beta = 1,376$$

$$\beta = 0,727$$

$$\beta = 1,376$$

$$H(z) = \frac{z-1}{z-1,376}$$

polo en 1,376
esto fuera del
radio unitario, es
Inestable

$$\beta = 0,727$$

$$H(z) = \frac{z-1}{z-0,727} = \frac{e^{j\Omega} - 1}{e^{j\Omega} - 0,727} = \frac{\cos \Omega - 1 + j \sin \Omega}{\cos \Omega - 0,727 + j \sin \Omega}$$

$$|H(\Omega)| = \frac{\sqrt{(\cos \Omega - 1)^2 + \sin^2 \Omega}}{\sqrt{(\cos \Omega - 0,727)^2 + \sin^2 \Omega}}$$

$$H(\Omega) = \sqrt{\frac{\cos^2 \Omega - 2 \cos \Omega + 1 + \sin^2 \Omega}{\cos^2 \Omega - 2 \cdot 0,727 \cos \Omega + 0,727^2 + \sin^2 \Omega}}$$

$$|H(\omega)| = \sqrt{\frac{2(1 - \cos(\omega))}{1,529 - 1,454 \cos(\omega)}}$$

$$|H(\omega = \pi)| = \sqrt{\frac{2(1 - (-1))}{1,529 - 1,454(-1)}} = \sqrt{\frac{4}{2,933}} = 1,158$$

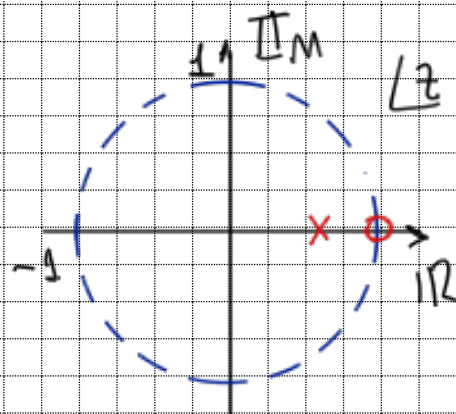
$$H(\omega = 0,1\pi) = \sqrt{\frac{2(1 - 0,951)}{1,529 - 1,454(0,951)}} = 0,818$$

$$\frac{0,818}{1,158} \approx 0,706 \approx \frac{\sqrt{2}}{2} \text{ verifica}$$

$$H(z) = \frac{z-1}{z-0,727}$$

polo em $z = 0,727$

cero em 1



$$|H(\omega)| = \sqrt{\frac{2(1 - \cos(\omega))}{1,529 - 1,454 \cos(\omega)}}$$

$$H(0) = 0 \quad |H(0,1\pi)| = 0,818 \quad |H(\pi)| = 1,15$$

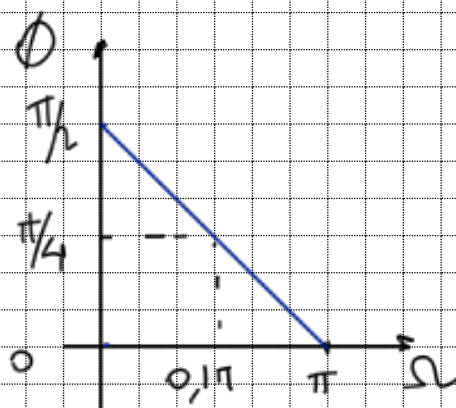
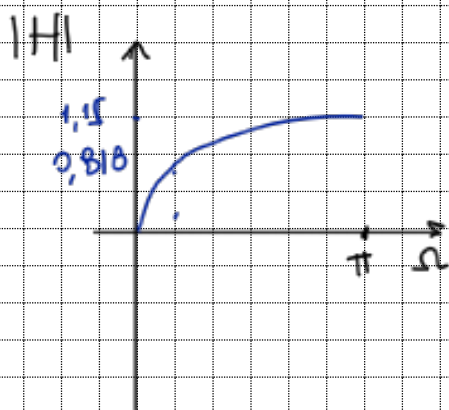
$$H(\omega) = \frac{\cos \omega - 1 + j \sin \omega}{\cos \omega - 0,727 + j \sin \omega}$$

$$\phi_H(\omega) = \tan^{-1}\left(\frac{\sin \omega}{\cos \omega - 1}\right) - \tan^{-1}\left(\frac{\sin \omega}{\cos \omega - 0,727}\right)$$

$$\phi(0) = \frac{\pi}{2}$$

$$\phi(0,1\pi) = 99^\circ - 54^\circ = 45^\circ$$

$$\phi(\pi) = 0$$



Filtro Peine

$$C_1 = -1 \quad N = 8$$

$$a_0 = b_0 = 1 \quad a_1 = a_2 = b_1 = b_2 = 0$$

$$H(z) = (1 - C_1 z^{-N}) \frac{b_0 + b_1 z^{-1} + b_2 z^{-2}}{a_0 + a_1 z^{-1} + a_2 z^{-2}}$$

$$H(z) = 1 - (-1)z^{-8}$$

$$H(z) = 1 + z^{-8}$$

$$H(z) = \frac{z^8 + 1}{z^8}$$

$$H(\omega) = 1 + e^{-j8\omega}$$

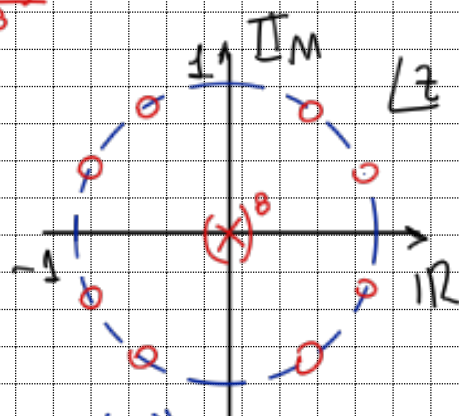
$$z^8 = -1$$

$$-1 = e^{j(2k+1)\pi} = z^8$$

$$z = e^{j\frac{(2k+1)\pi}{8}}$$

$$\text{polos: } \left\{ \frac{1}{8}\pi, \frac{3}{8}\pi, \frac{5}{8}\pi, \right.$$

$$\left. \frac{7}{8}\pi, \frac{9}{8}\pi, \frac{11}{8}\pi, \frac{13}{8}\pi, \frac{15}{8}\pi \right\}$$



$$H(\omega) = e^{-j4\omega} (e^{j4\omega} + e^{-j4\omega}) = e^{-j4\omega} (2 \cos(4\omega))$$

$$H(\omega) = 2 \cos(4\omega) e^{-j4\omega}$$

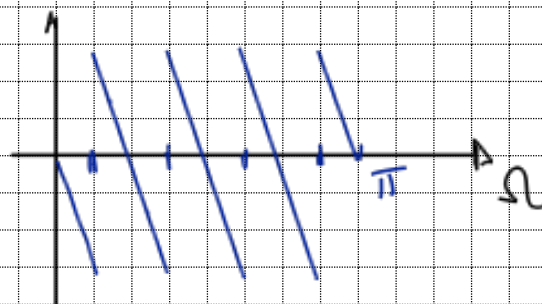
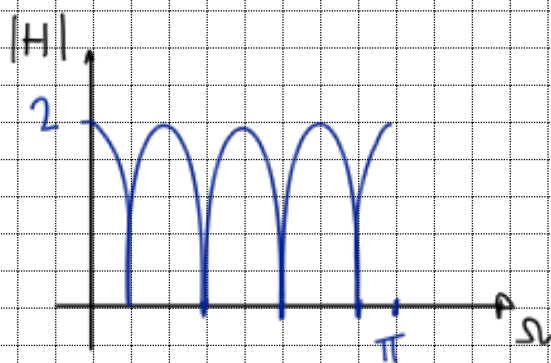
$$|H(\omega)| = 2 |\cos(4\omega)| \Rightarrow \cos 4\omega = 0 \quad \text{en } \omega = \frac{1}{8}\pi, \frac{3}{8}\pi$$

$$\Rightarrow \left(\frac{2k+1}{8} \right) \pi \Rightarrow \frac{1}{8}\pi, \frac{3}{8}\pi, \frac{5}{8}\pi, \frac{7}{8}\pi$$

$$\phi(\omega) = \tan^{-1} \left(\frac{-\sin(8\omega)}{(\cos(8\omega) + 1)} \right) \quad \text{en } \frac{\pi}{8}, \frac{3\pi}{8}, \frac{5\pi}{8}, \frac{7\pi}{8} \text{ indefinido}$$

$$\phi(0) = 0 \quad \phi\left(\frac{\pi}{16}\right) = -45^\circ \quad \phi\left(\frac{3\pi}{16}\right) = 45^\circ \quad \phi\left(\frac{5\pi}{16}\right) = -45^\circ \quad \phi\left(\frac{7\pi}{16}\right) = 45^\circ$$

$$\phi\left(\frac{9\pi}{16}\right) = -45^\circ \quad \phi\left(\frac{11\pi}{16}\right) = 45^\circ \quad \phi\left(\frac{13\pi}{16}\right) = -45^\circ \quad \phi\left(\frac{15\pi}{16}\right) = 45^\circ$$



- $$H(\Omega) = 2 \cos 4\Omega e^{-j4\Omega}$$

$$|H(\Omega)| = 2 |\cos(4\Omega)|$$

PEAKS:

$$|\cos(4\Omega)| = 1 \quad / \quad \Omega = 0; \pi; 2\pi; 3\pi; \dots; k\pi$$

$$4\Omega = k\pi \Rightarrow \Omega = \frac{k\pi}{4}$$

$$\Omega = 0, \Omega = \frac{\pi}{4}, \Omega = \frac{\pi}{2}, \Omega = \pi$$

NULLS:

$$|\cos(4\Omega)| = 0 \quad / \quad \Omega = \frac{\pi}{8}, \frac{3\pi}{8}, \dots, \frac{(2k+1)\pi}{8}$$

$$\Omega = \frac{(2k+1)\pi}{8}$$

$$\Omega = 2\pi \frac{f}{f_s} \Rightarrow f = \frac{\Omega}{2\pi} f_s$$

$$f_s = 8 \text{ kHz}$$

$$\left\{ \begin{array}{l} \text{Peaks: } \frac{k\pi}{4} \frac{f_s}{2\pi} = \frac{k}{8} f_s \end{array} \right.$$

$$\left\{ \begin{array}{l} \text{Nulls: } \frac{(2k+1)\pi}{8} \frac{f_s}{2\pi} = \frac{(2k+1)}{16} f_s \end{array} \right.$$

k	0	1	2	3	4
Peaks	0	1000	2000	3000	4000
Nulls	500	1500	2500	3500	4500

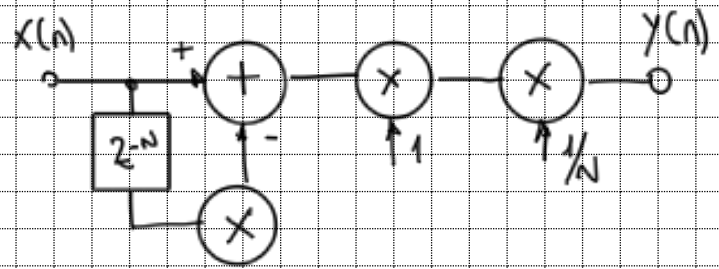
$4500 > \frac{f_s}{2}$
 equivale a 3500 Hz
 por aliasing

Diagrama de bloques

① Filtro medio m.v.l

$$H(z) = \frac{1}{N} \frac{1 - z^{-N}}{1 - z^{-1}}$$

$$N=3, 1, 5$$



② Diferenciador

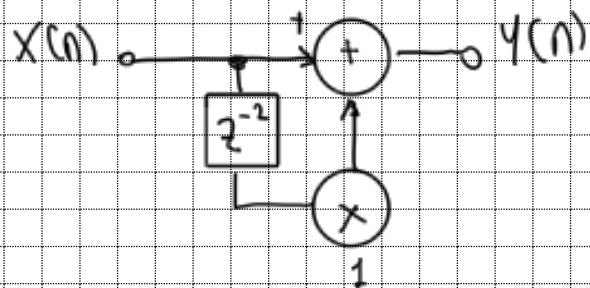
$$H(z) = 1 - \frac{z}{2}$$

$$b_0 = 1/2$$

$$b_1 = b_2 = 0$$

$$a_0 = 1 \quad a_1 = a_2 = 0$$

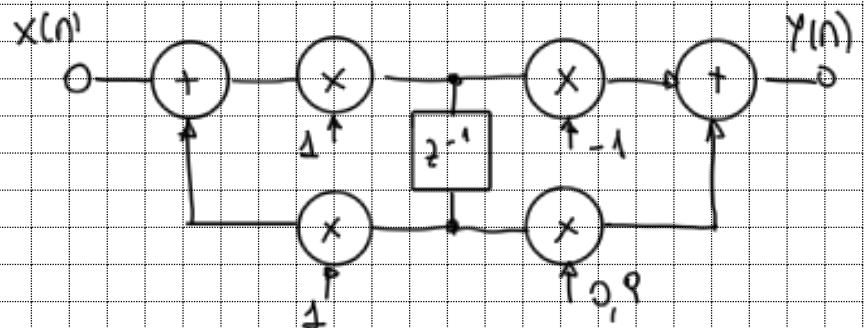
$$c_1 = 1 \quad N = 2$$



③ DC Blocker

$$H(z) = \frac{z-1}{z-0,9}$$

$$c_1 = b_2 = a_2 = 0$$



④ Peine

$$H(z) = 1 + z^{-8}$$

$$N = 8$$

